

PROJECT DOCUMENTATION

Strategic Employee Attrition Intelligence Dashboard

Date	21-09-2026
Team ID	MCA
Project Name	Strategic Employee Attrition Intelligence Dashboard
Maximum Marks	Project Documentation

This document records the complete project work step by step — from problem definition and planning, through data collection, quality assessment, exploration and preprocessing, business questions and visualisation design, dashboard and report development, to final insights and recommendations.

1. Define Problem Statement

Employee attrition is a major challenge for organizations because frequent employee turnover can increase recruitment costs, training expenses, workload on existing employees, and productivity losses. However, organizations often struggle to identify the key factors that cause employees to leave.

The problem is to analyze employee data and determine the factors associated with employee attrition, examining aspects such as job role, department, salary, job satisfaction, overtime, work-life balance, years of experience, and other employee characteristics to identify patterns among employees who leave. The goal is to generate data-driven insights that help HR teams understand attrition drivers, identify high-risk employee groups, and support effective employee retention strategies.

1.1 Customer Problem Statements

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Priya Nair, Chief People Officer	Understand how department, job role, work mode & AI adoption impact attrition	I can't easily tell which depts/roles are most at risk	Attrition data is scattered across systems	Uncertain when setting retention budgets; unable to design targeted interventions
PS-2	Marcus Webb, People Analytics Manager	Analyze how overtime, burnout, manager support & work-life balance influence attrition	I can't spot early warning signs of disengagement	No unified way to flag teams at risk of burnout-driven attrition	Reactive instead of proactive; worried interventions come too late
PS-3	Dr. Sofia Alvarez, Workforce	Analyze how salary hike %, years since	I struggle to pinpoint which factors are the	Compensation, promotion & performance	Unable to give HR leadership confident,

	Research Lead	promotion, performance & training affect attrition risk	strongest predictors at each career stage	data are not analyzed together	evidence-based recommendations
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2. Project Proposal (Proposed Solution)

2.1 Project Overview

Objective	Quantify overall and segment-level attrition rates (department, job role, work mode); identify the strongest predictors of attrition (burnout, overtime, compensation growth, tenure, performance); build an interactive Power BI dashboard for HR and leadership decision-making; provide AI-assisted insights for root-cause analysis; and deliver actionable recommendations to reduce attrition and improve workforce wellbeing.
Scope	Covers analysis of all 5,000 employee records — demographics, compensation, tenure, workload, engagement, performance, and attrition outcomes — including segment-level breakdowns by department, job role, and work mode, and delivery of an interactive Power BI dashboard and report.

2.2 Problem Statement

Description	895 of the organization's 5,000 employees (17.9%) have left, and HR cannot currently pinpoint which factors — compensation, workload, engagement, career progression, or work arrangement — most strongly predict attrition.
Impact	Without this understanding, HR cannot prioritize retention spend or design targeted interventions, which drives up recruitment and training costs, increases workload on remaining employees, and reduces overall productivity.

2.3 Proposed Solution Approach

- Analyze employee demographics, compensation, tenure, workload, engagement, performance, and AI adoption attributes against the attrition outcome.
- Build an interactive Power BI dashboard with KPI cards, attrition-rate charts by department/job role/work mode, decomposition tree for multi-dimensional drill-down, and filters for dynamic exploration.
- Develop complementary reports focused on retention drivers (burnout, work-life balance, salary growth, tenure) and performance/career growth (overtime intensity, performance ratings, training investment, marital status, age).
- Deliver actionable, evidence-based recommendations for HR and leadership to reduce attrition and improve workforce wellbeing.

3. Initial Project Planning — Product Backlog & Sprint Schedule

The project was executed in three sprints covering data collection & preparation, KPI and visualisation development, and dashboard integration & testing.

Sprint	Epic	ID	User Story / Task	Points	Priority	Start	End
Sprint-1	Data Collection & Preparation	DS-1	As an HR manager, I want to collect employee data from multiple sources and consolidate it into a single dataset for analysis.	3	High	15-09-2026	21-09-2026

Sprint-1	Data Cleaning	DS-2	As a data analyst, I want to clean and preprocess the data (handle missing values, duplicates and outliers) so that it is ready for visualization and analysis.	3	High	16-09-2026	21-09-2026
Sprint-2	KPI Development	DS-3	As an HR manager, I want to calculate key metrics (attrition rate, retention rate, average tenure, etc.) so that I can track overall workforce health.	3	Medium	16-09-2026	21-09-2026
Sprint-2	Attrition Analysis Visualizations	DS-4	As a business leader, I want to create visualizations (attrition by department, job role, and satisfaction) so that I can identify key trends and risk factors.	5	High	17-09-2026	21-09-2026
Sprint-3	Dashboard Integration & Testing	DS-5	As a stakeholder, I want to integrate the visualizations into an interactive dashboard and test the functionality so that I can use it for decision-making.	5	Medium	18-10-2026	21-09-2026

4. Data Collection Plan & Raw Data Sources

4.1 Project Overview (Data Context)

The project analyzes a 5,000-record employee HR dataset to quantify overall and segment-level attrition (17.9%, 895 employees), identify the strongest predictors of attrition (burnout, overtime, compensation growth, tenure, performance), and build an interactive Power BI dashboard with AI-assisted insights for HR and leadership decision-making.

4.2 Data Collection Plan

Data was collected as a single structured HR dataset (CSV) containing employee-level records with 29 attributes per employee — demographics, job details, compensation, tenure, workload, engagement, performance, AI tool adoption, and the attrition outcome — loaded directly into Power BI for cleaning, transformation, and analysis.

4.3 Raw Data Sources Identified

Source Name	Description	Location / Format	Size	Access	Notes
Employee HR Dataset	Employee-level HR data: demographics, compensation, tenure, workload,	employee_attrition_dataset.csv (CSV)	5,000 × 29	Private (internal)	Primary source

	engagement, performance, AI usage, attrition (29 attributes).				
Dept & Job Role Lookup	Dimension tables for department, job_role, work_mode categories.	Derived via Power Query	< 1 MB	Private (internal)	Supporting lookups

5. Data Quality Report

The following data quality issues were identified and resolved systematically prior to analysis.

Data Source	Data Quality Issue	Severity	Resolution Plan
Employee Dataset	Missing values in monthly_income, burnout_score, engagement_score	Moderate	Imputed missing numeric values using department/job-role median in Power Query; flagged records for HR follow-up.
Employee Dataset	Duplicate employee_id records	High	Removed duplicates via Power Query Remove Duplicates, retaining the most recently updated record per employee_id.
Employee Dataset	Inconsistent category labels (department, work_mode, marital_status)	Moderate	Standardized category values using a mapping/lookup table during Power Query transformation.
Employee Dataset	Outlier values in commute_minutes and overtime_hours_per_week	Moderate	Reviewed outliers against business-realistic ranges; corrected or removed invalid entries before model load.
Employee Dataset	Incorrect data types (uses_ai_tools_at_work as text; salary_hike_pct with % symbol)	Low	Converted to Boolean and Decimal during Power Query Change Type step.
Employee Dataset	Invalid values in years_since_promotion and num_prior_companies (e.g. negatives)	High	Validated against years_at_company and total_working_years; corrected or removed failing records.
Employee Dataset	Inconsistent attrition labels ('Y'/'N' mixed with 'Yes'/'No')	Low	Standardized attrition values to Yes/No using Power Query Replace Values.

6. Data Exploration and Preprocessing

6.1 Data Overview

5,000 employee records with 29 attributes covering: demographics (age, gender, marital status, education level); job details (department, job role, job level); compensation (monthly income, stock option level, salary hike %); tenure (years at company, total working years, years since promotion, prior companies); workload (commute minutes, overtime hours/week, business travel days/year, training hours); wellbeing/engagement (manager support, burnout, engagement, work-life balance scores); performance (performance rating, last review score); AI adoption (uses_ai_tools_at_work,

perceived_ai_job_risk); and the target variable attrition (Yes/No) — 895 employees (17.9%) have attrition = Yes.

6.2 Data Cleaning

- Checked all 29 columns for missing values and imputed or removed them where appropriate (e.g. monthly_income, burnout_score, engagement_score).
- Removed duplicate employee_id records.
- Standardized inconsistent category labels in department, job_role, work_mode, and marital_status.
- Validated that attrition and uses_ai_tools_at_work contain only valid Yes/No and True/False values.

6.3 Data Transformation

- Used Power Query to filter out invalid or blank rows, sort records by department and job_role.
- Pivoted data to compute segment-level attrition rates by department, job role, and work mode.
- Added calculated columns such as tenure bands, income bands, and a burnout/engagement risk flag.

6.4 Data Type Conversion & Modelling

- Set employee_id to text; numeric fields to whole number/decimal; uses_ai_tools_at_work to True/False; categorical fields to text categories.
- Defined a single Employee fact table keyed on employee_id, with relationships to lookup tables for Department, Job Role, and Work Mode.
- Built DAX measures for Attrition Rate (%), Average Burnout Score, Average Engagement Score, Average Tenure, and Average Salary Hike %, each segmented by department, job role, and work mode.
- Saved the cleaned and transformed dataset within the Power BI data model (.pbix) and exported a backup cleaned CSV.

7. Business Questions and Visualisation Report

Visualization development is the process of representing employee attrition data through graphical and interactive visuals. Power BI was used to transform employee-related data into meaningful visualizations that help identify patterns and factors associated with employee attrition.

7.1 Business Questions & Matching Visuals

1. What is the overall employee attrition rate? — KPI Card + Donut Chart (fields: attrition, employee_id). Purpose: understand overall proportion of employees who left.
2. Which departments have the highest employee attrition? — Clustered Column / Horizontal Bar Chart (department, attrition). Purpose: identify high-turnover departments.
3. How does employee attrition vary across different job roles? — Bar Chart (job_role, attrition). Purpose: compare attrition across roles.
4. What is the relationship between employee engagement / burnout and attrition? — Column / Scatter / Clustered Bar (engagement_score / burnout_score, attrition). Purpose: examine wellbeing drivers of turnover.
5. How does overtime workload affect employee attrition? — Bar Chart by overtime band (overtime_hours, attrition). Purpose: quantify risk of high overtime intensity.
6. How does attrition vary by work mode (Onsite / Hybrid / Remote)? — Bar Chart (work_mode, attrition). Purpose: assess work-arrangement impact.

7. What is the impact of AI tool usage on attrition? — Clustered Bar (uses_ai_tools_at_work, attrition count). Purpose: explore AI-adoption relationship with retention.
8. How does attrition change across tenure / career stages? — Line Chart (years_at_company / age / job_level, attrition rate). Purpose: identify risk windows in the employee journey.

8. Dashboard Design

Creating an effective dashboard involves thoughtful design to ensure that the presented information is clear, relevant, and easily understandable for the intended audience.

8.1 Design Principles Applied

- Clear and Intuitive Layout – KPI cards, charts, and filters organized for easy navigation and quick insight discovery.
- Use Appropriate Visualizations – Horizontal bar charts for attrition rates, clustered bar for AI impact, hierarchical decomposition tree for multi-dimensional analysis.
- Colour and Theming – Consistent blue colour palette with clear contrast for readability and professional appearance.
- Interactive Filters and Slicers – Dropdown filters for Work Mode, Department, and Job Role enable dynamic exploration.
- Drill-Down Capabilities – Decomposition tree allows progressive drilling from department → job role → work mode → job level → overtime → burnout score.
- Responsive Design – Layout optimized for both desktop and mobile viewing within Power BI.
- Custom Visuals and Icons – KPI cards with clear metric labels and a decomposition tree for hierarchical insights.
- Use of Infographics – Visual storytelling through combined charts that highlight attrition drivers at a glance.

8.2 Dashboard Screenshot

The following screenshot presents the fully interactive Strategic Employee Attrition Intelligence Dashboard built in Power BI:

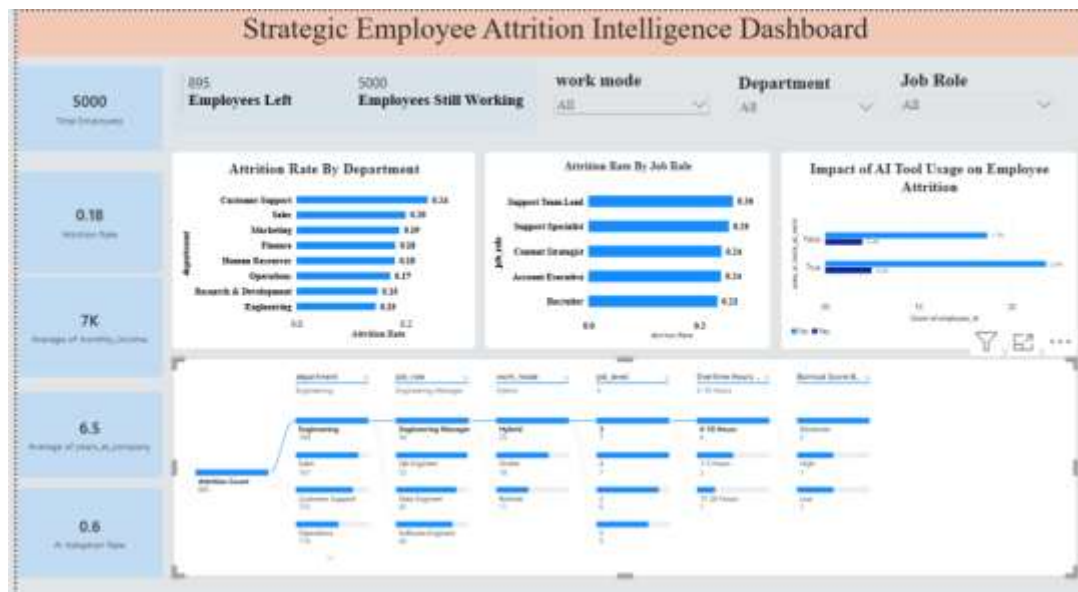


Figure 1: Strategic Employee Attrition Intelligence Dashboard (Power BI)

8.3 Key Dashboard Insights

- 5,000 employees analyzed; 895 employees left; overall attrition rate $\approx$ 18%.

- Customer Support has the highest departmental attrition; Support Team Lead and Support Specialist roles show elevated attrition.
- Average monthly income $\approx$ 7K; average tenure $\approx$ 6.5 years; AI adoption rate $\approx$ 60%.
- Filters for work mode, department, job role, overtime, and burnout enable deeper multi-dimensional analysis.

9. Report Design

A report is a comprehensive document that provides a detailed and structured account of data analysis, findings, and insights. Two complementary reports were developed to support in-depth exploration of retention drivers and performance/career growth factors.

9.1 Key Observation Categories

1. Trends Over Time – Analyze how attrition rate, engagement, and burnout change over tenure bands to identify risk windows.
2. Performance Comparisons – Compare attrition across departments, job levels, work modes, and education backgrounds.
3. Employee Segmentation – Assess demographic and workplace data (age, marital status, education, overtime intensity) to tailor retention strategies.
4. Goal Achievement – Measure progress against KPIs (attrition rate, engagement index, manager support, burnout score).

9.2 Report 1: Employee Retention Performance Intelligence

This report focuses on retention drivers including burnout, work-life balance, engagement, manager support, salary growth, education, work mode, tenure, and career level. Interactive filters for Education Level and Work Mode enable targeted exploration.



Figure 2: Employee Retention Performance Intelligence Report

9.3 Key Insights – Retention Performance

1. Highest Attrition Rate: Employees with 18.10 years at company recorded attrition rate of 1.00, followed by 16.10 and 15.10 years.
2. Lowest Attrition Rate: Employees with 11.60 years recorded the lowest rate at 0.04.
3. Wide Variation: Across all 146 years-at-company values, Attrition Rate ranged from 0.04 to 1.00.

4. Education: Employees with a Bachelor's degree had the highest Attrition Rate at 0.19 (2,562 employees).
5. Work Arrangement & Burnout: Hybrid employees with a Very High Burnout Score contributed 18.89% of the Attrition Rate.
6. Burnout Impact: Very High Burnout category had the highest average Attrition Rate at 0.49, followed by High, Moderate, and Low.

9.4 Report 2: Employee Performance Career Growth Intelligence

This report examines performance ratings, manager support, age, work experience, overtime intensity, training investment, career stages, marital status, and department-wise attrition. Filters for Gender and Marital Status support demographic slicing.



Figure 3: Employee Performance Career Growth Intelligence Report

9.5 Key Insights – Performance & Career Growth

1. **Headline Metrics:** Avg Performance Rating 3.62 | Avg Manager Support 6.61 | Avg Age 37.12 | Avg Work Experience 11.70 years | Avg Weekly Overtime 6.83 hours.
2. **Attrition Risk by Overtime Intensity:** Climbs from 0.12 (1–5 hrs/week) to 0.54 (21–30 hrs/week).
3. **Attrition Risk Across Performance Levels:** Rating 2 shows attrition 0.29 vs Rating 5 at 0.08.
4. **Attrition Risk by Burnout / Training:** Falls from 0.54 (Very High burnout) to 0.05 (Low burnout).
5. **Career Stages (by Age):** Relatively stable through most stages with a pronounced spike around the late-50s.
6. **Marital Status:** Married employees form the largest share of both retained and departed staff, followed by Single and Divorced.
7. **Department-Wise Attrition:** Customer Support 16.51%, Sales 13.73%, Operations 12.44%; Engineering and R&D lowest.

10. Summary and Recommendations

The Strategic Employee Attrition Intelligence Dashboard and supporting reports provide a multi-dimensional view of workforce risk and opportunity. Key findings include:

- Overall attrition of approximately 18% (895 of 5,000 employees).
- Customer Support and Support-role positions as primary departmental/role risk areas.
- Elevated attrition among long-tenure employees, Bachelor's degree holders, hybrid workers with very high burnout, high-overtime cohorts, and lower-rated performers.

- Burnout and overtime intensity emerge as strong, actionable retention levers.

10.1 Recommended Actions

- Prioritize burnout mitigation and workload rebalancing for Hybrid employees and high-overtime bands.
- Design targeted retention programs for Customer Support and Support Team Lead / Specialist roles.
- Review compensation growth and promotion pathways for mid-to-late career employees and Bachelor's degree cohorts.
- Strengthen manager support and performance development for lower-rated performers.
- Continue monitoring via the interactive dashboard and reports for early warning signs and continuous improvement.

This documentation consolidates the complete project lifecycle — problem definition, planning, data collection & quality, exploration & preprocessing, business questions, visualisation, dashboard and report design — into a single professional record of the Strategic Employee Attrition Intelligence Dashboard project.

10.2 Git Hub Link :

Git repository URL : <https://github.com/sanamehabooba/Strategic-Employee-Attrition-Intelligence-Dashboard>